Important Notes

1. All geometry was adjusted to standard slicer settings and basic PLA, you can add walls and infill for stronger parts, using PETG will also enforce parts.

2. In most cases flat bolts can be swapped to countersunk bolts.

3. We used TPU by bambulab for tires, they should print in 7H, standard settings for TPU in bambulab slicer are recommended.

4. Main grey color used in videos is light grey 10104 PLA by bambu.

5. We highly recommend using 2F modules after base and before roof, when you are using modifications.

6. Holes for casters are M8, make sure to check it before purchasing them.

Notes: 1. All geometry tuned to standard slicer & PLA; add walls/infill for strength. 2. TPU recommended for tires. 3. Highly recommend 2F modules after base & before roof.

VAULT



3D Printed Modular Storage System

Instructions Manual

Hardware List

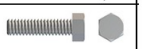
DRAWERS

1.	M4 14MM bolts	x4	*per floor*	
2.	M6 50MM bolts	x2	*per roof handle*	

SUSTAINERS

1.	M4 10MM bolts	x8	*per module*	
2.	M4 14MM bolts	x16	*per module*	
3.	M4 20MM bolts	x7	*per module*	

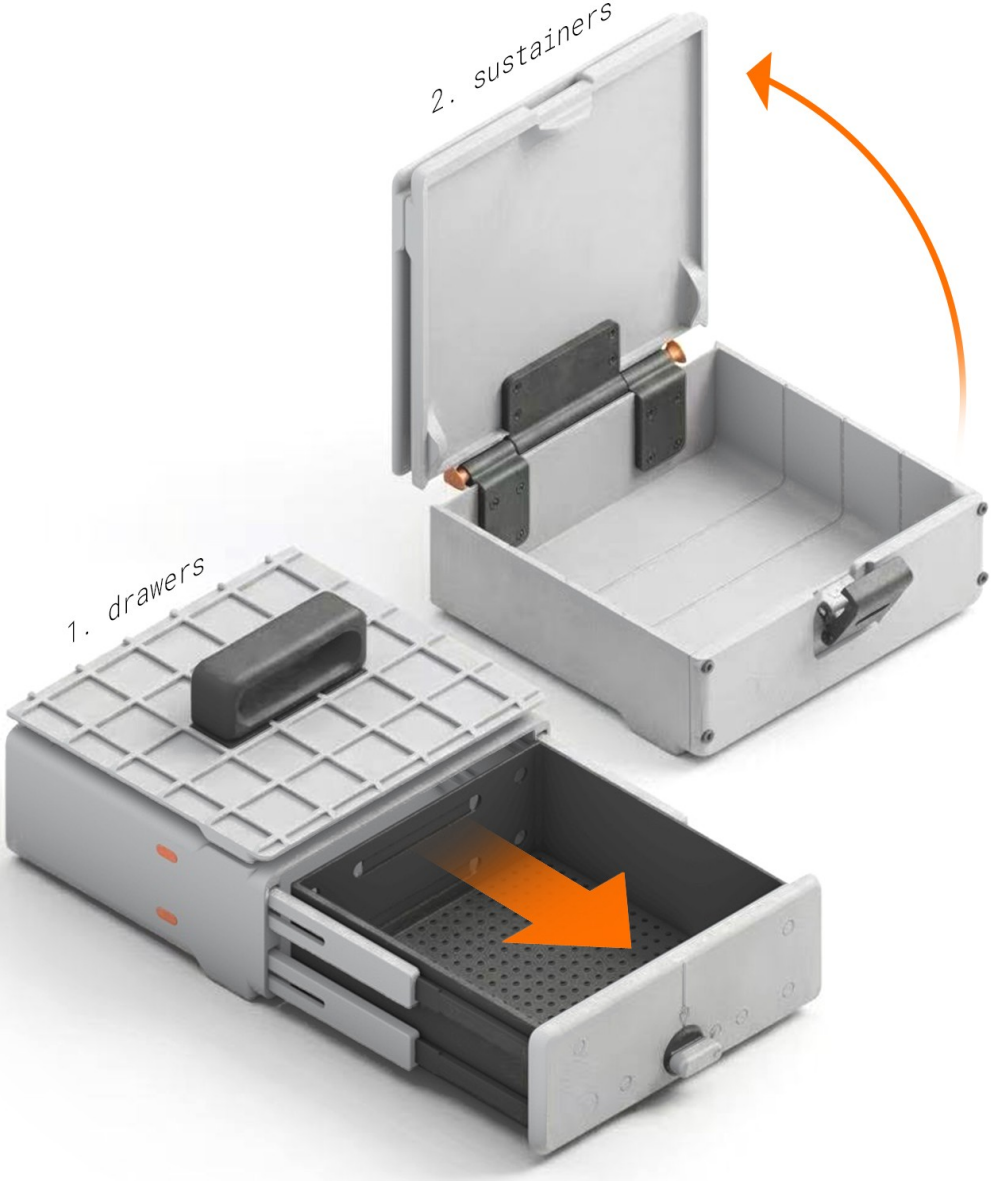
MODIFICATIONS

1.	M6 30MM bolts	x4	*per pull-out handle*	
2.	M6 35MM bolts	x2	*per pull-out handle*	
3.	M6x8x8 heat insert	x2	*per pull-out handle*	
4.	M4 60MM bolts	x8	*per static handle*	
5.	M4 40MM bolts	x4	*per pegboard board*	
6.	M4 60-100MM bolts	x4	*per backbone board*	
7.	M16 504MM rod	x1	*per XL base*	
8.	M16 414MM rod	x1	*per L base*	
9.	M16 338MM rod	x1	*per M base*	
10.	M16 248MM rod	x1	*per S base*	
11.	M16 N-MM rod	x2	*per pull-out handle*	
12.	M4 25MM bolts	x6	*per wheel*	
13.	M6 90MM bolts	x2	*per foot*	
14.	6205ZZ bearing	x1	*per wheel*	
15.	SCSAW stopper	x1	*per wheel*	

Five Ways of Using VAULT



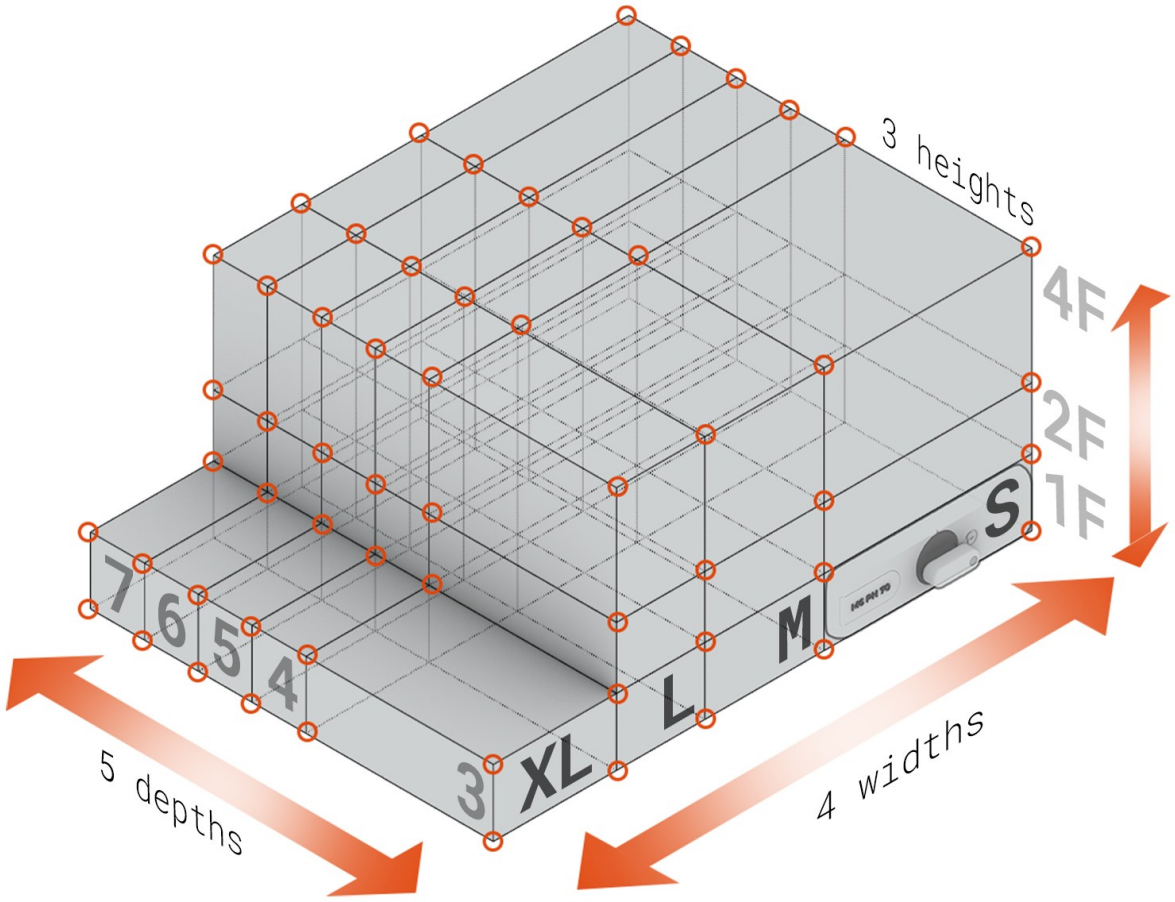
Two Types of VAULT: Drawers & Sustainers



* Archive is split into two parts: Drawers & Sustainers. Choose according to your setup.

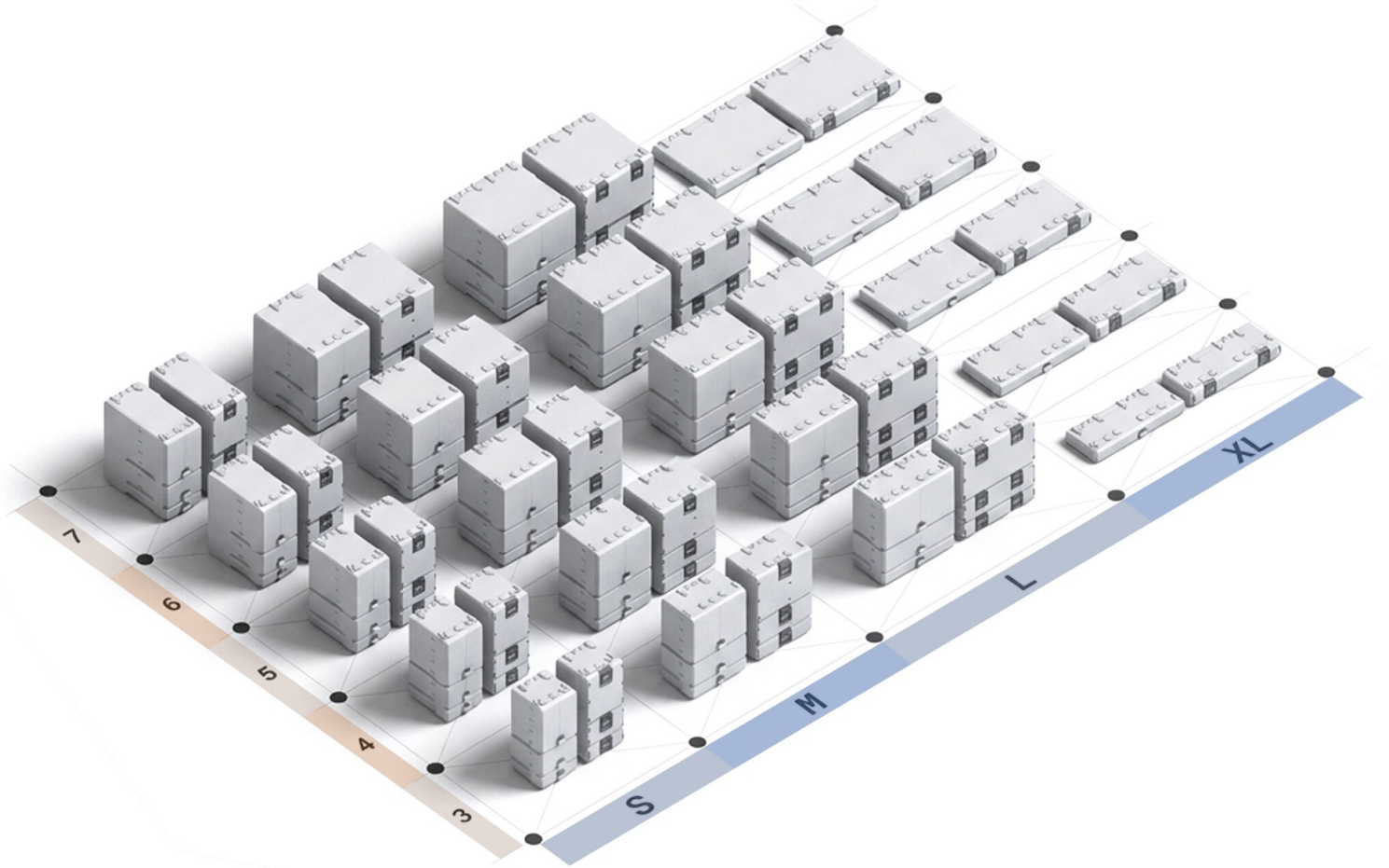
*

S, M, L, XL — 50 Size Variants

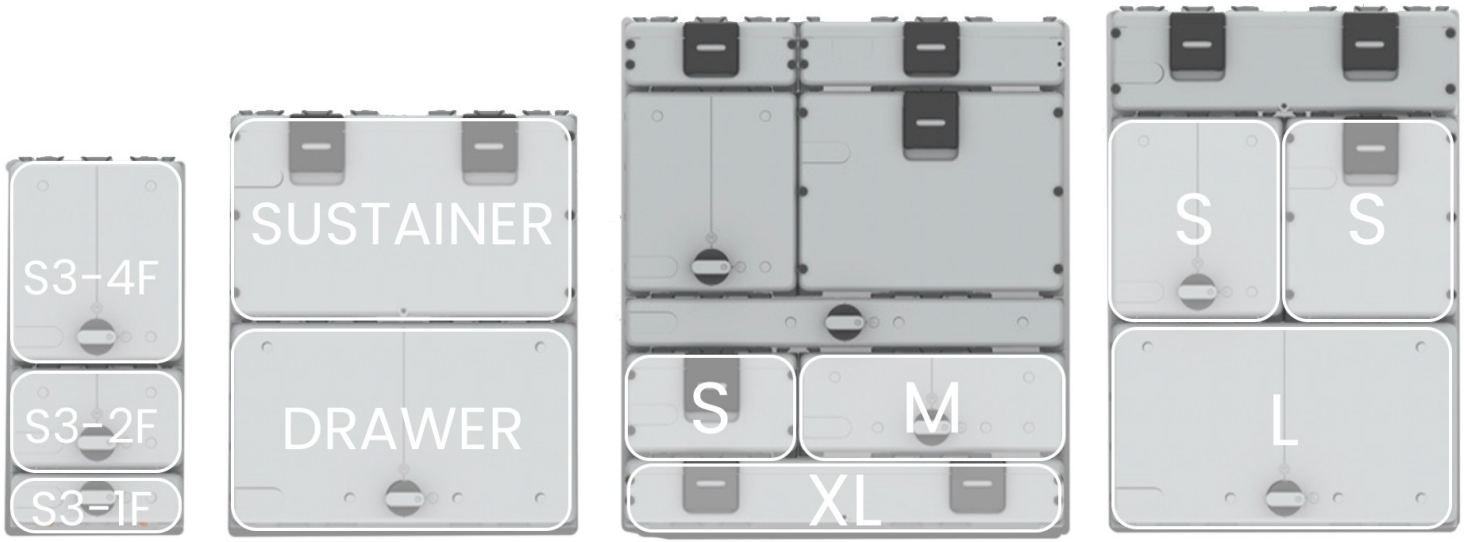


* * Depths & widths determined by Gridfinity units (42mm) & bed size; heights include 1F, 2F, 4F (no 3F to avoid gaps; everything divides by 2).

160 Module Variants Overview

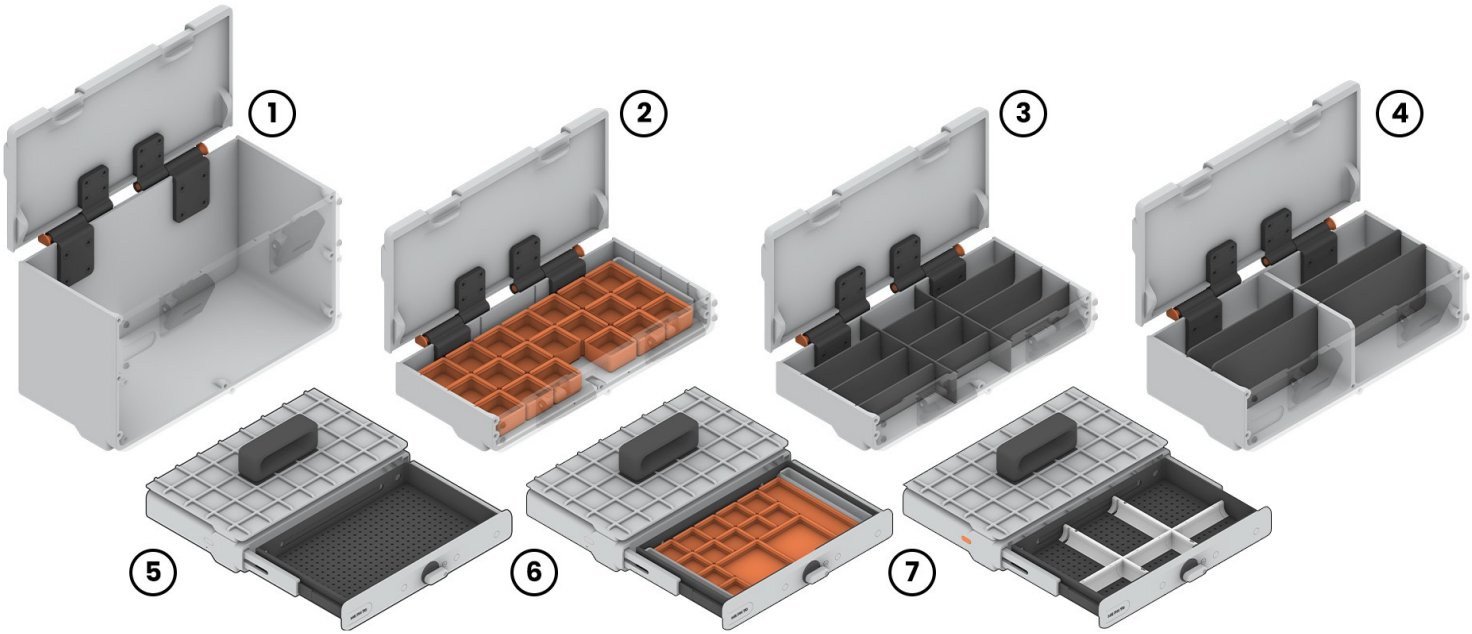


Stacking & Combination Rules



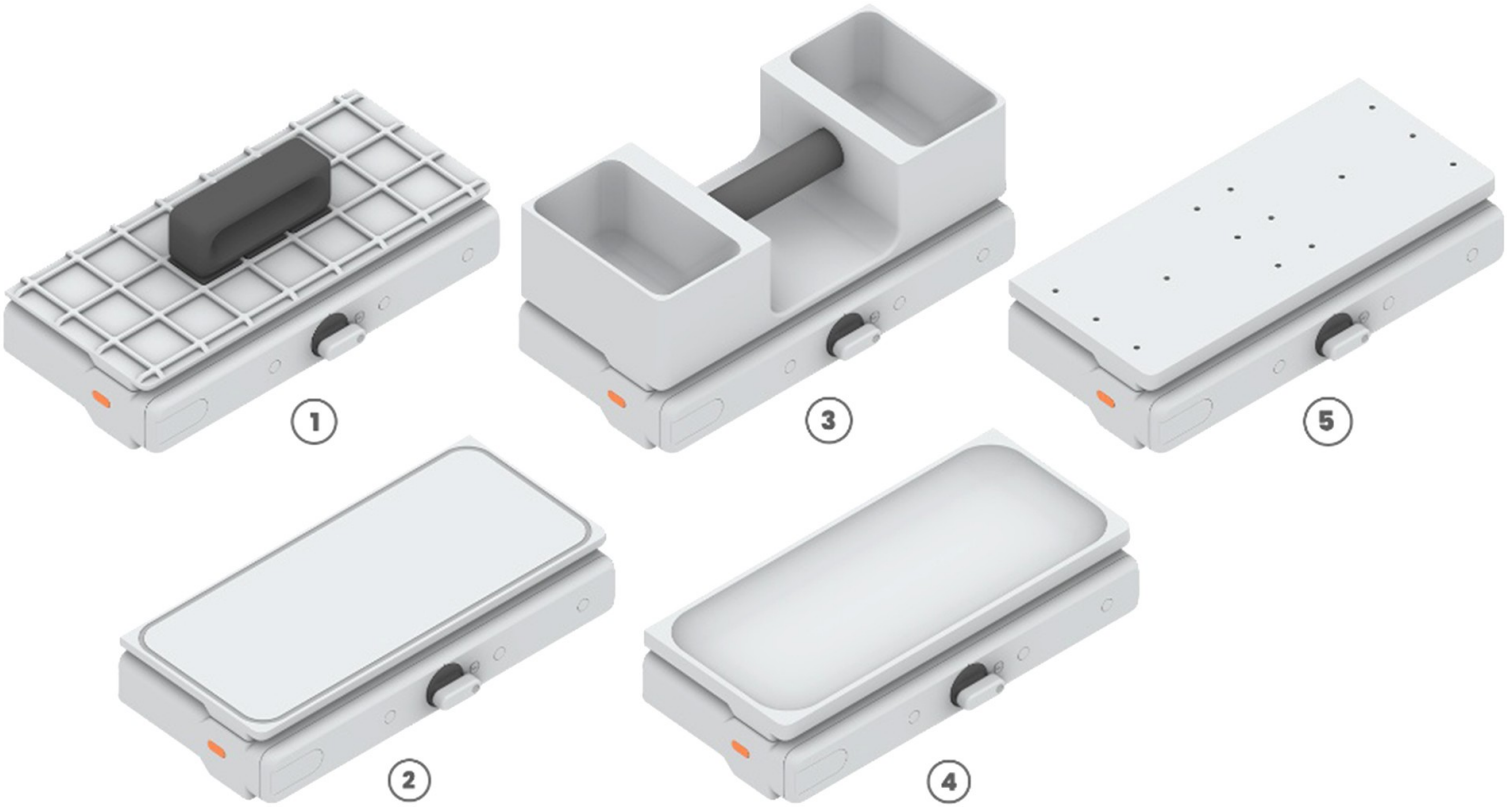
1. Modules with similar depth and width stack on each other.
2. SUstainers and Drawers can stack as well

1. Modules of similar depth & width stack. 2. Sustainers & Drawers mix-stack. 3. XL = S + M stacks evenly. 4. L = S + S stacks seamlessly.



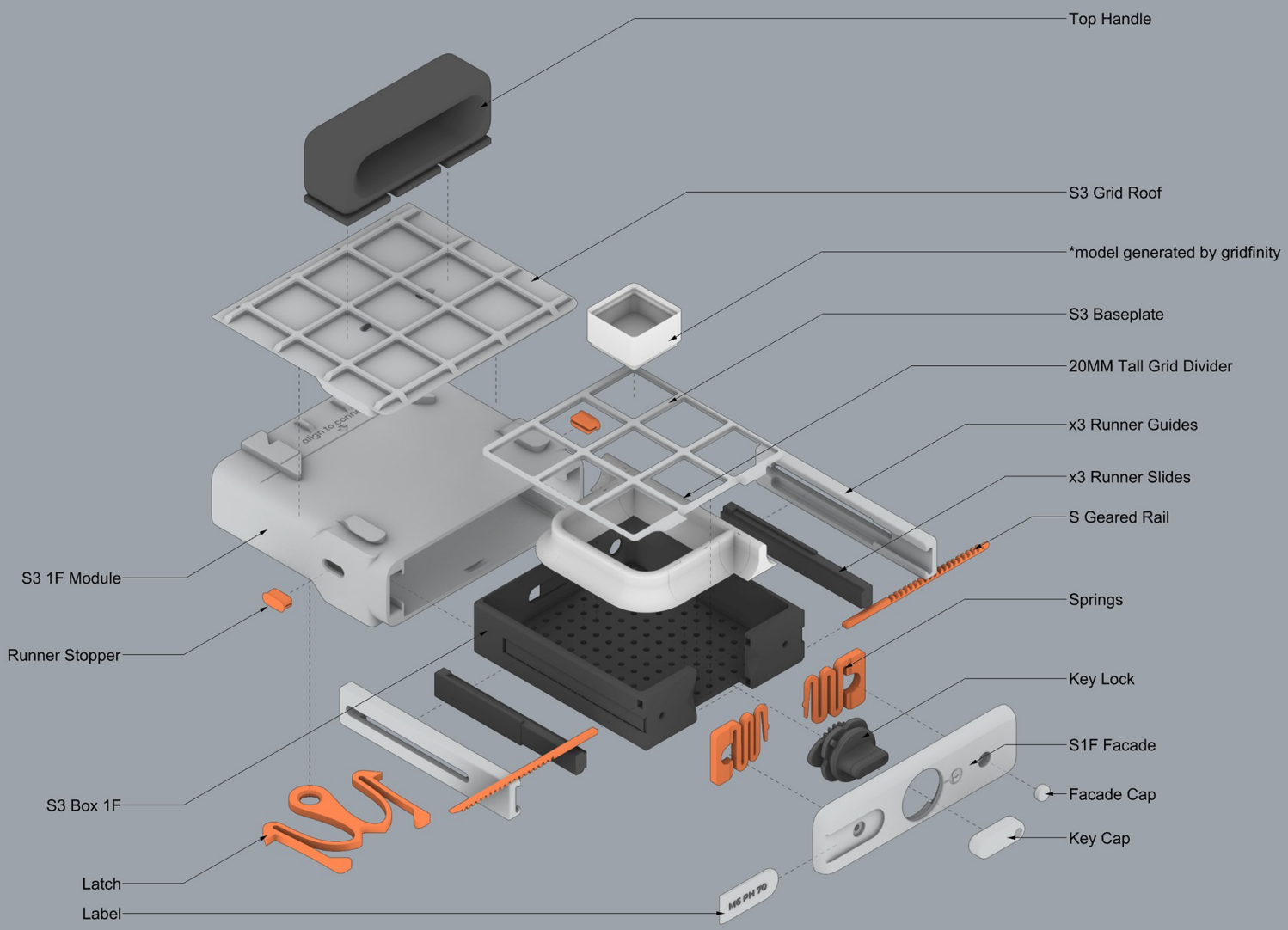
1. Flat Sustainer 2. Gridfinity Base 3. Empty Drawer 4. Sealed Sustainer 5. Dividers Sustainer 6. Gridfinity Base 7. Grid Dividers

Roof & Top Panel Types

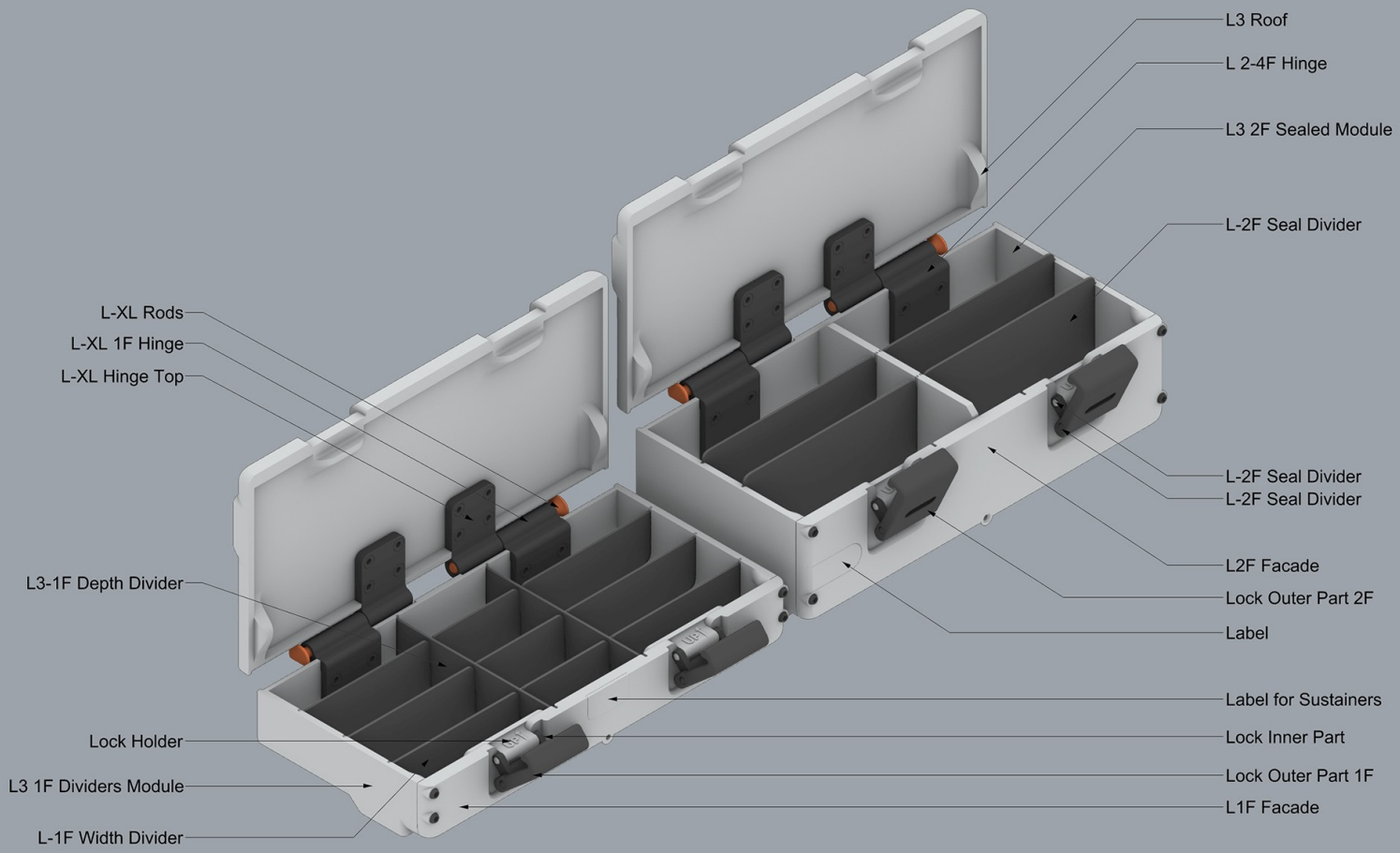


1. Flat Roof 2. Gridfinity Roof 3. Heavyweight Roof Insert 4. Bowl Roof 5. Under-Table Mount Roof

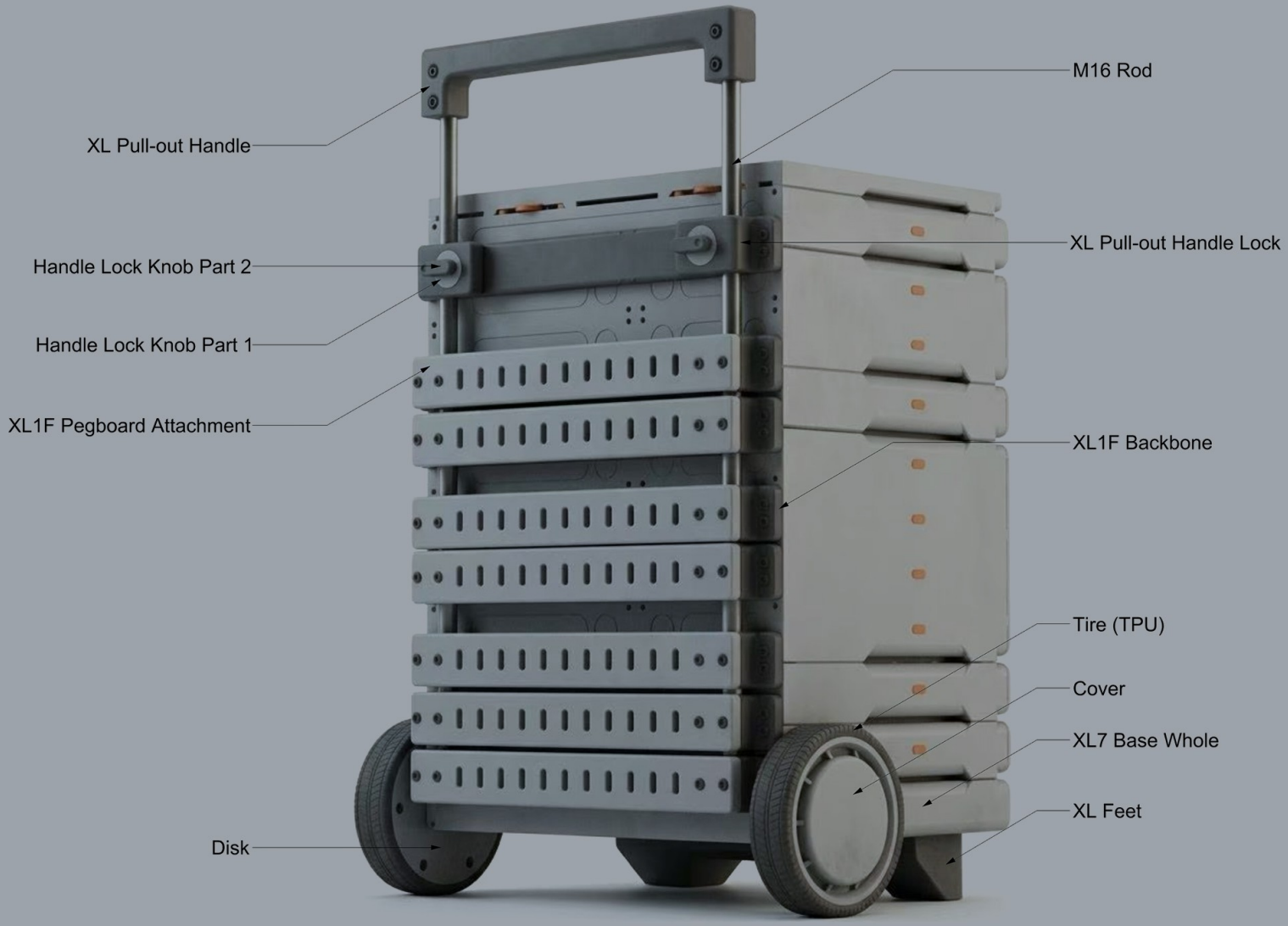
Drawers Parts Naming



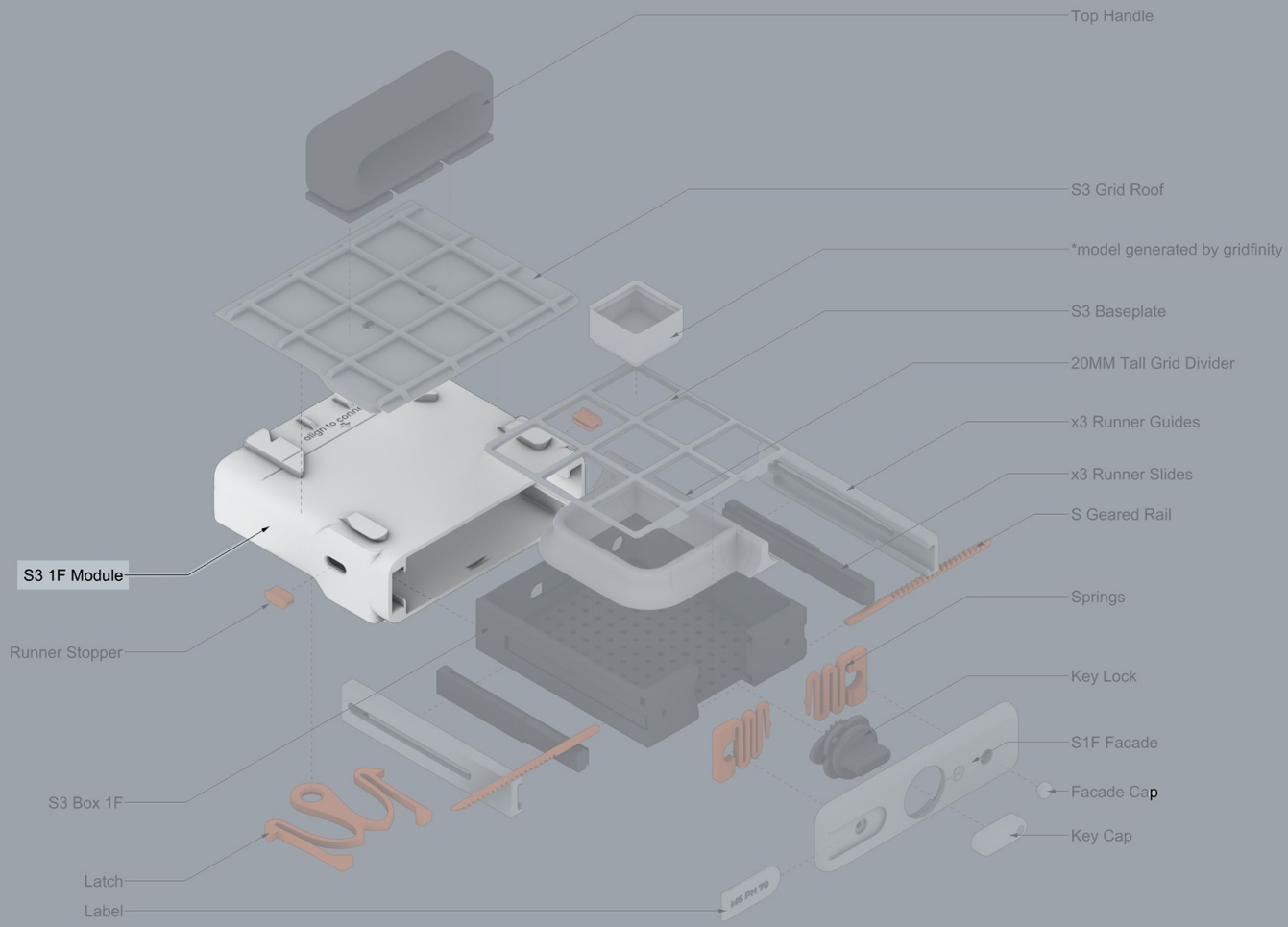
Sustainers Parts Naming

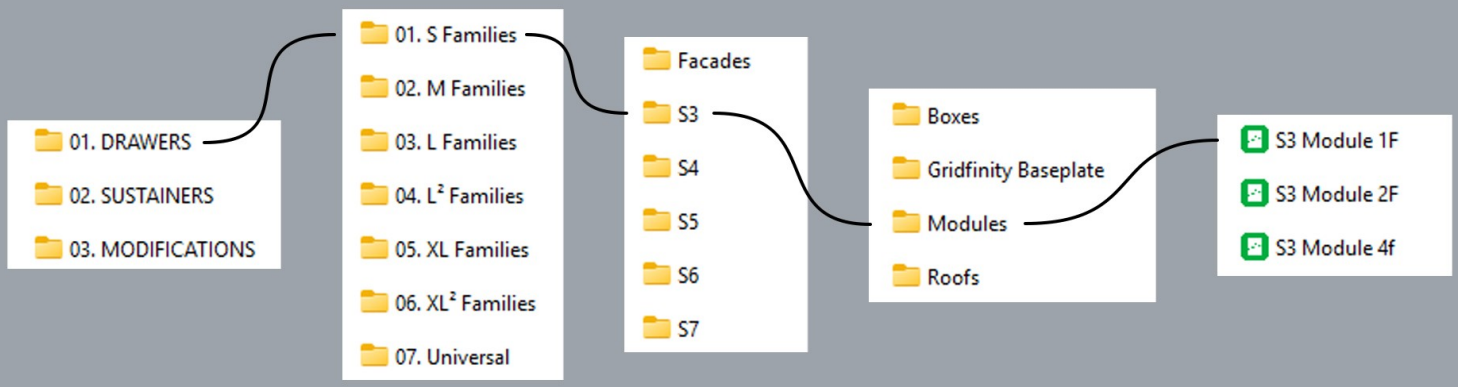


Modifications Parts Naming

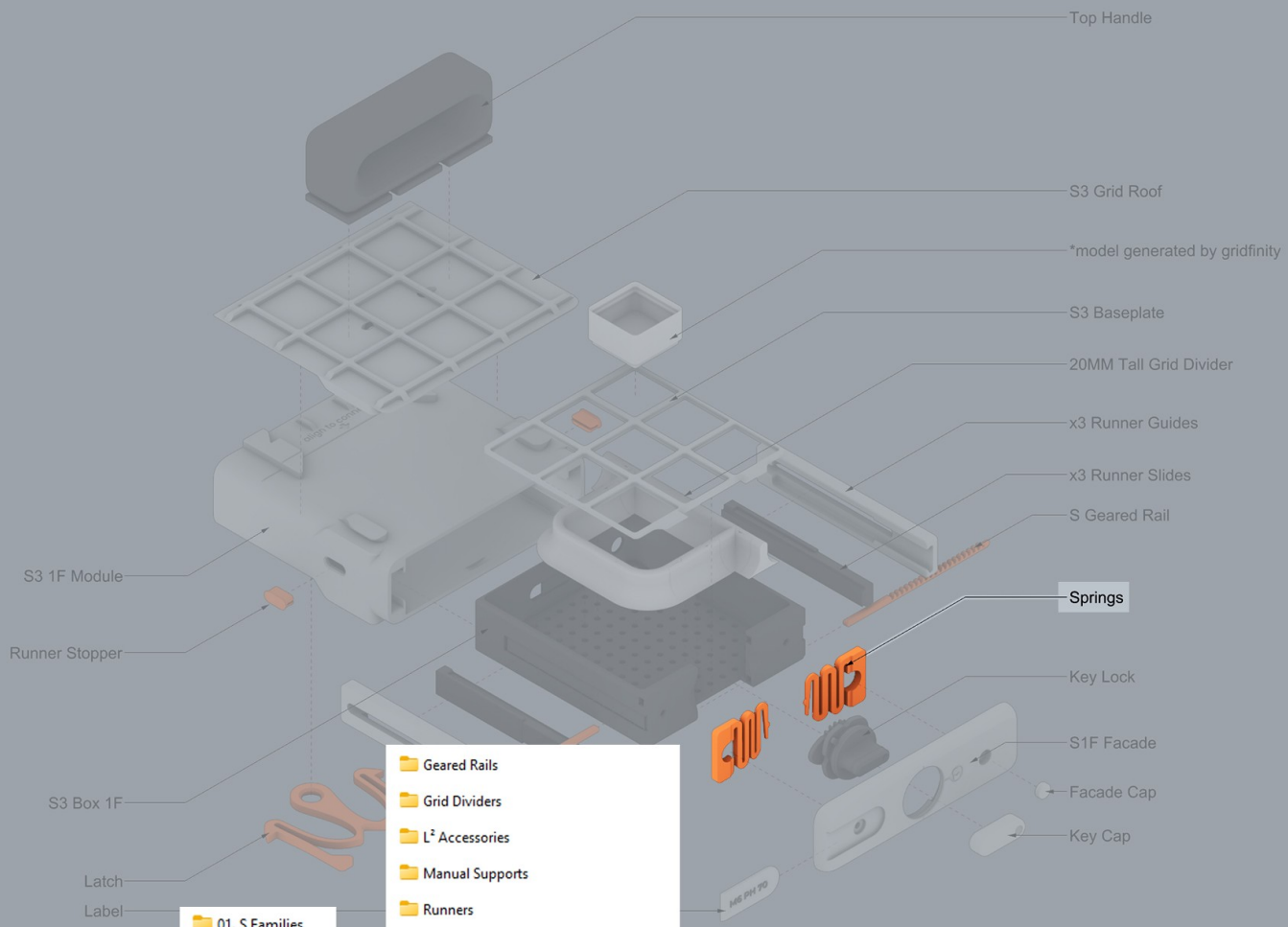


Archive Guide (1): Drawers & Facades





Archive Guide (2): Rails & Accessories

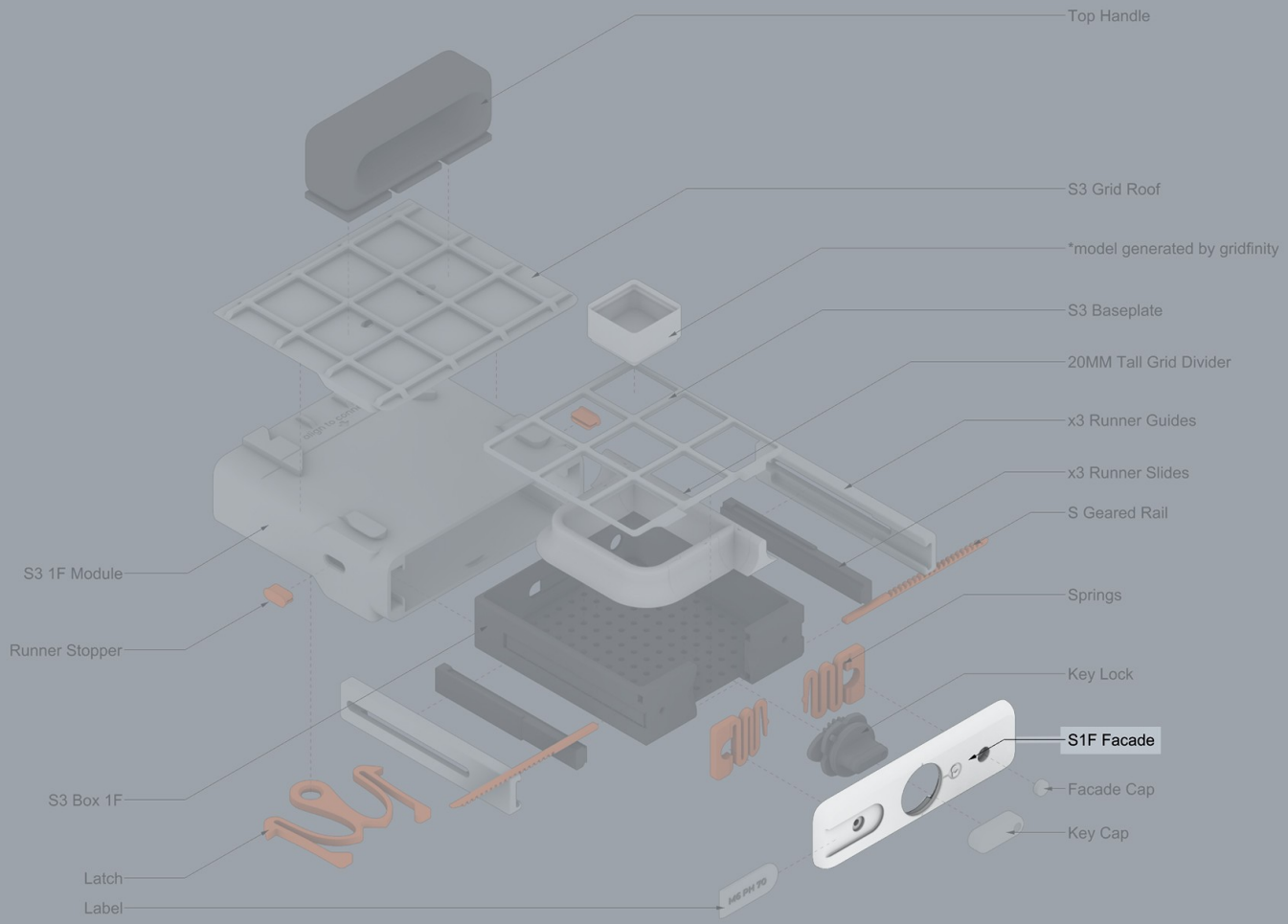


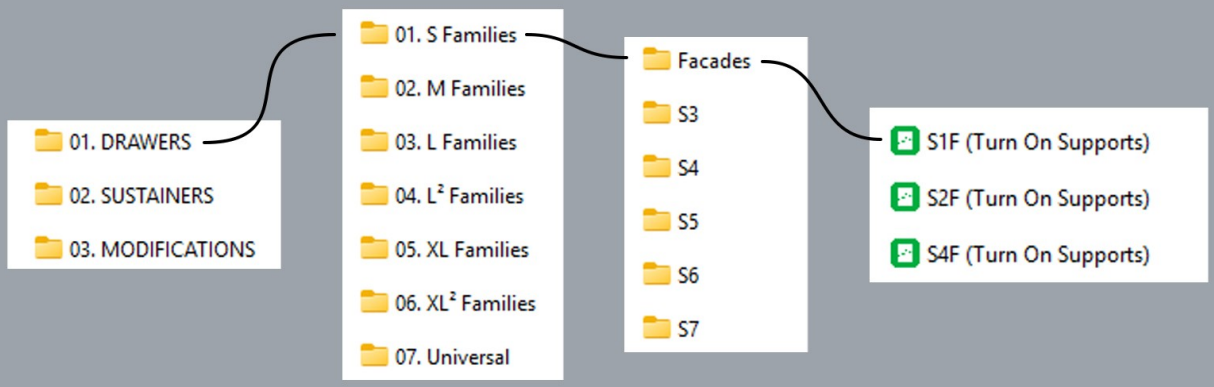
- 01. DRAWERS
- 02. SUSTAINERS
- 03. MODIFICATIONS

- 01. S Families
- 02. M Families
- 03. L Families
- 04. L² Families
- 05. XL Families
- 06. XL² Families
- 07. Universal

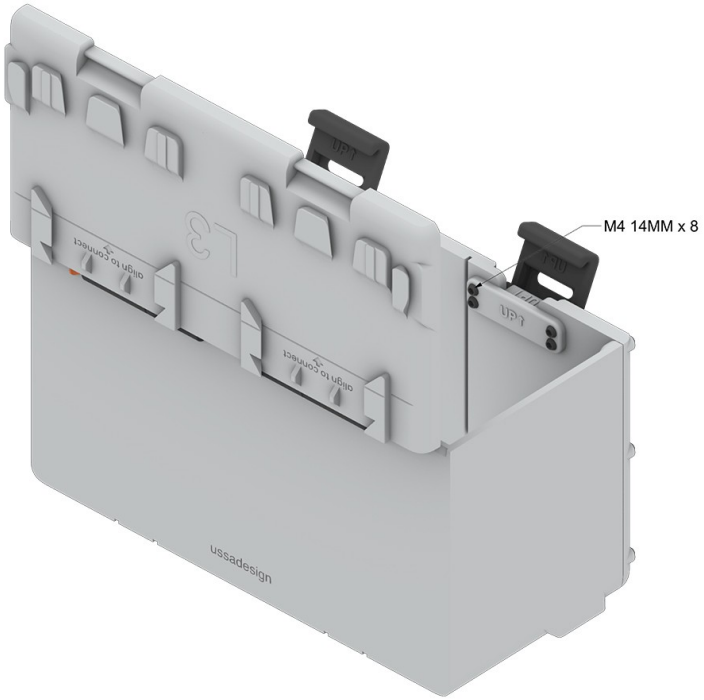
- Geared Rails
- Grid Dividers
- L² Accessories
- Manual Supports
- Runners
- Facade Cap
- Heawyweight Roof Handle
- Heawyweight Roof Insert
- Hidden Key (Turn On Supports)
- Hidden Key Part 2 (Turn on Supports)
- Key (Turn On Supports)
- Key Cap
- Label
- Latch
- Springs
- Top Handle (Turn On Supports)

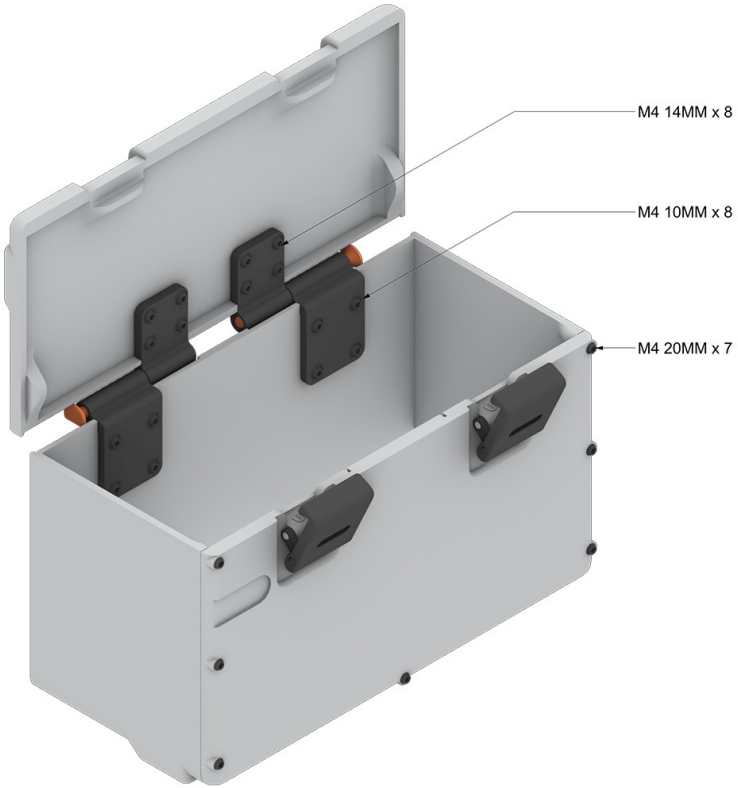
Archive Guide (3): Supports Settings





Required Hardware — Sustainers

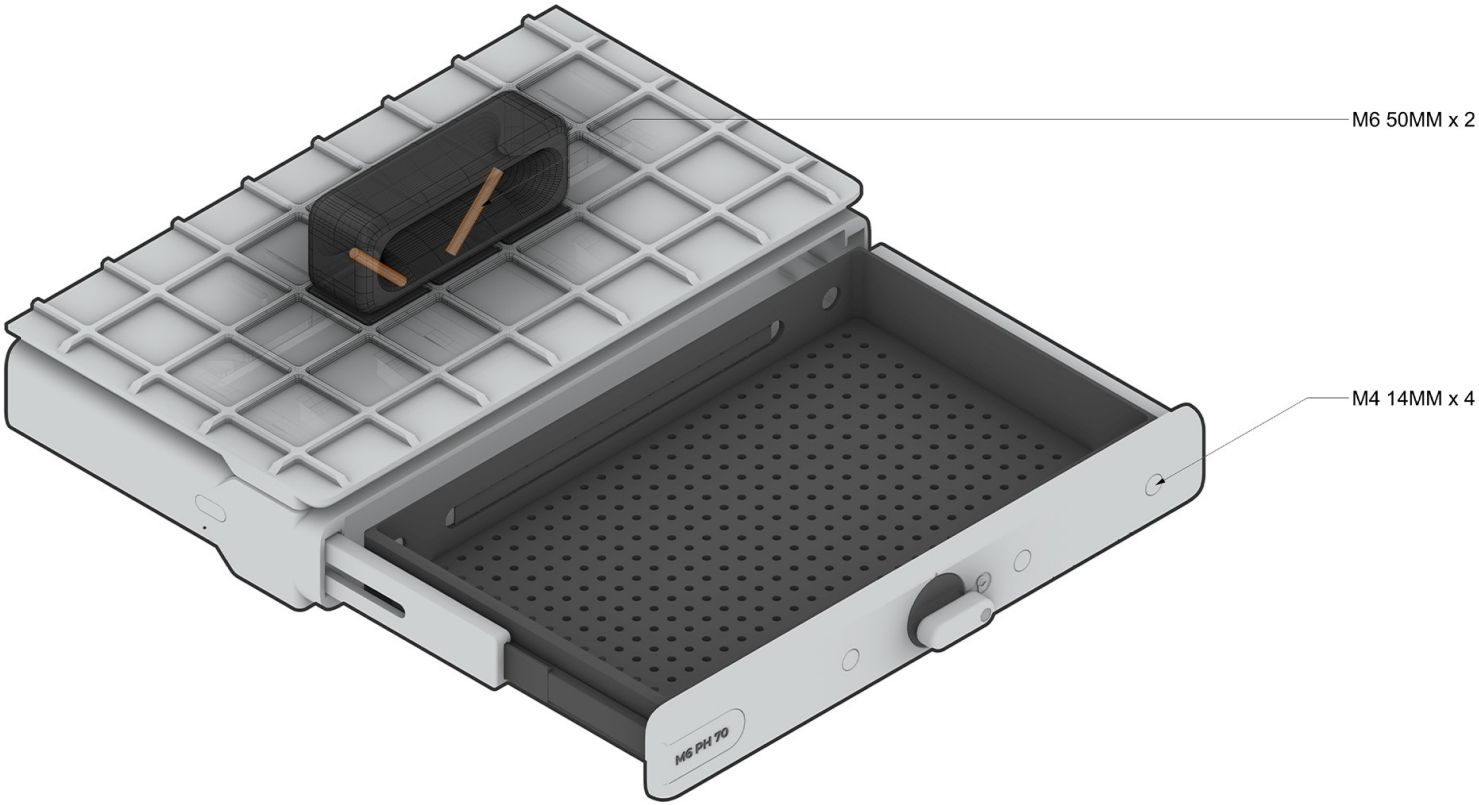




* Sustainers per module: 16× M4 14mm bolts, 8× M4 10mm bolts, 7× M4 20mm bolts.

*

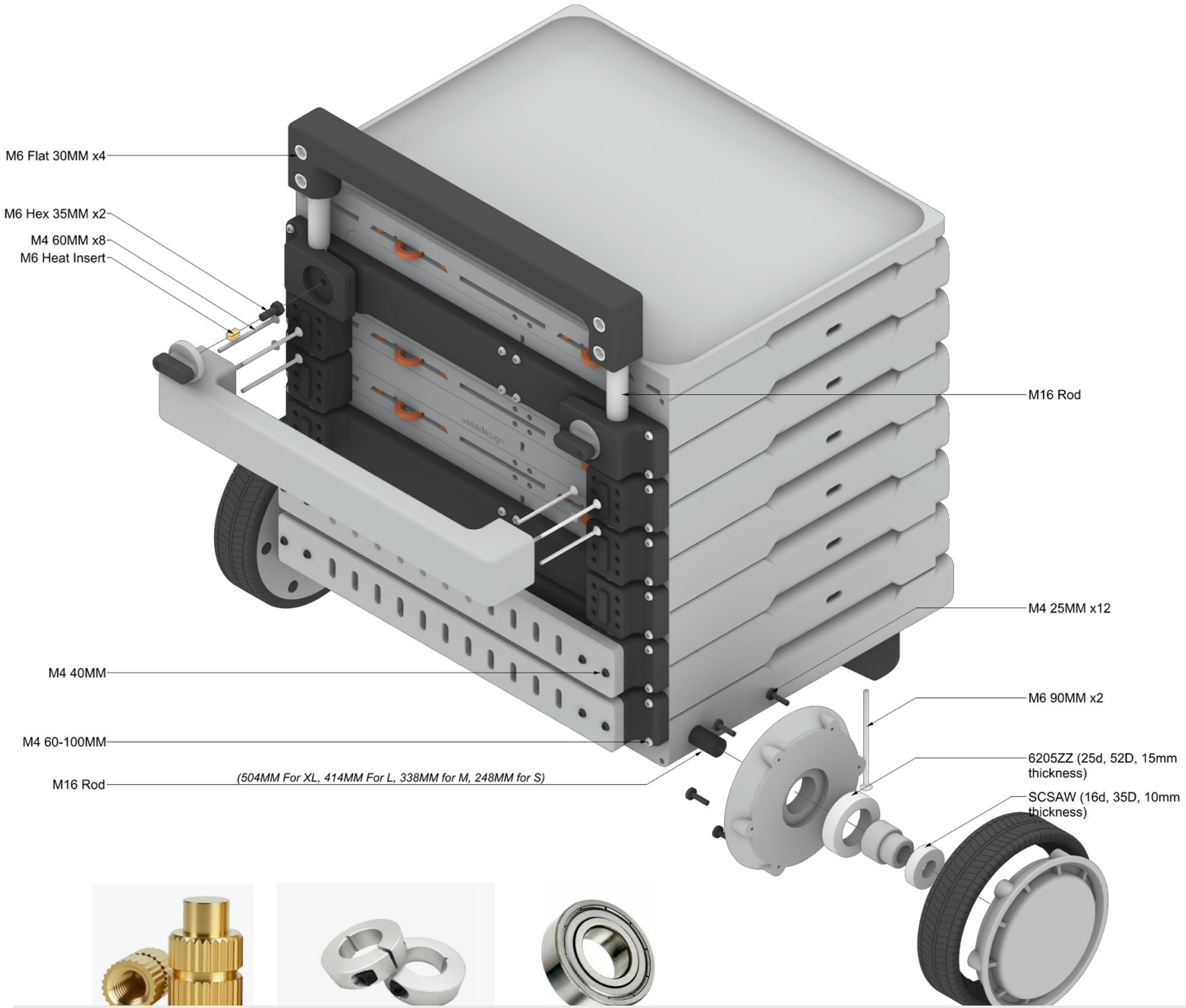
Required Hardware — Drawers



* Drawers: 4× M4 14mm bolts for 1F (6× for 2F/4F); 2× M6 50mm bolts per drawer.

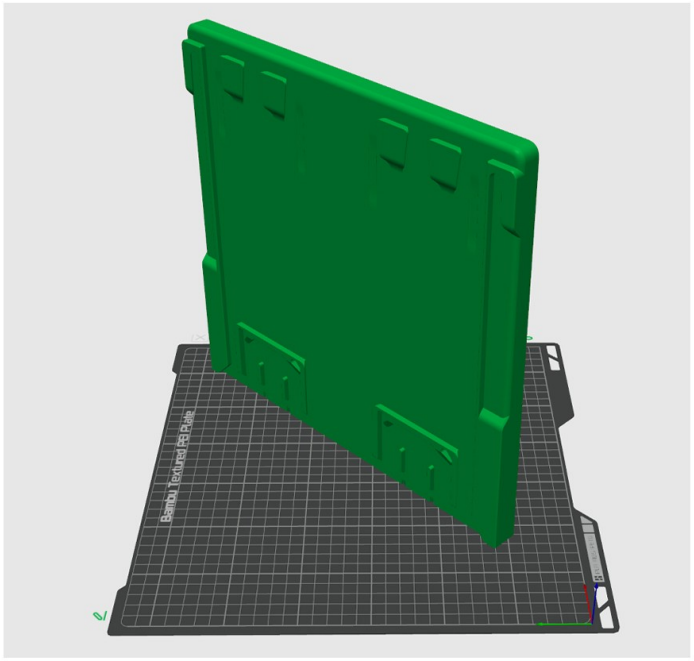
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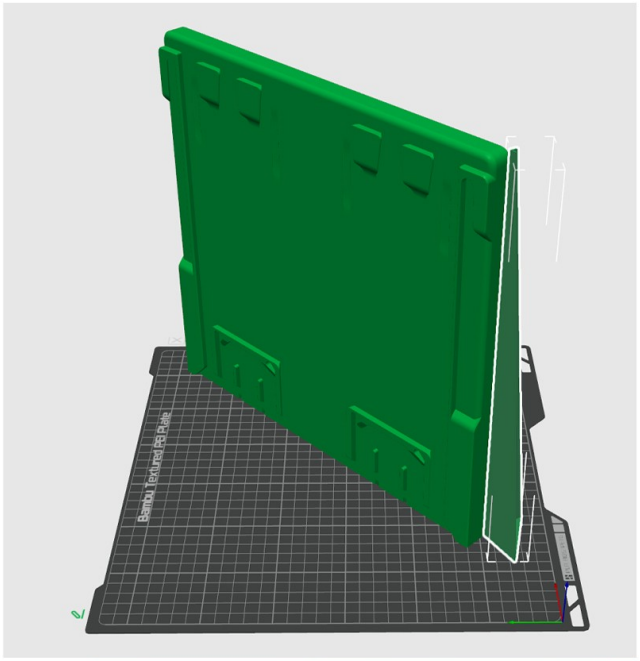
Required Hardware — Backbone & Wheels

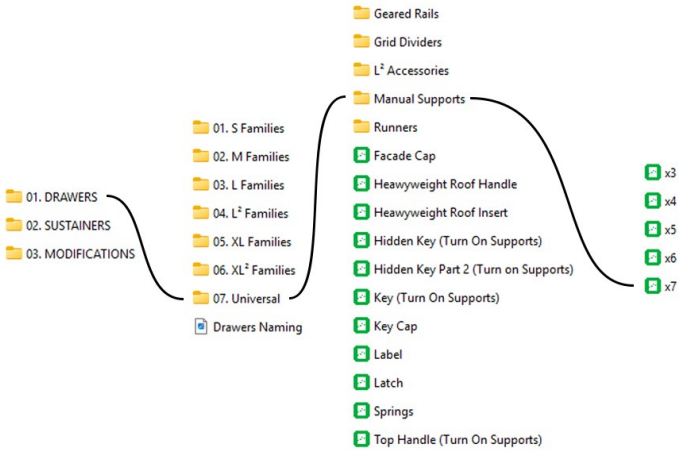


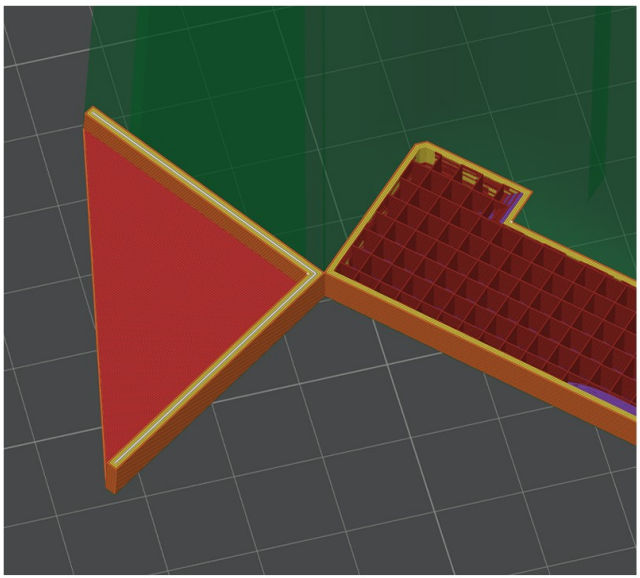
* Backbone & Casters: M6 Flat 30mm ×4, M6 Hex 35mm ×2, M4 60mm ×8, M6 Heat Insert, M16 Rod.

Manual Supports Guidelines









You can add manual supports for a more stable print experience, especially for tall prints.